

The empirical recurrence for OEIS A234110: recovering a conjecture that is not written down

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Abstract

OEIS A234110 records “Empirical recurrence of order 96”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 3125$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 96, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 96 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234110 is “Number of $(n+1) \times (4+1)$ 0..4 arrays with every 2×2 subblock having the sum of the absolute values of all six edge and diagonal differences equal to 12.”. It has offset 1 and begins

2064, 7742, 31740, 151304, 738628, 3840512, 20547980, 114726620, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 09:11:21, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 3125$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 3125$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 6250$ exact terms of the model returns order 96 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	12	c_{25}	1051923183725926029	c_{49}	1709615130755456705646929	c_{73}	2617
c_2	151	c_{26}	1215885939855941584	c_{50}	3003664339489630531143509	c_{74}	722
c_3	−2159	c_{27}	−7358961478237834900	c_{51}	−2469604398923171953683307	c_{75}	−668
c_4	−10782	c_{28}	−9239141260035823257	c_{52}	−4187174294746101871458921	c_{76}	−124
c_5	186630	c_{29}	44410141167530707134	c_{53}	3165964716166440447772070	c_{77}	138
c_6	489988	c_{30}	59999619108113717848	c_{54}	5092380071198416457633888	c_{78}	150
c_7	−10309178	c_{31}	−232097410700751654589	c_{55}	−3599813680666720928174611	c_{79}	−23
c_8	−16280140	c_{32}	−334476539272940156900	c_{56}	−5391972941914249976352044	c_{80}	−9
c_9	408166746	c_{33}	1054138086744970158579	c_{57}	3624261027964676031026645	c_{81}	29
c_{10}	436483471	c_{34}	1607162586965048660479	c_{58}	4956255423149915835248546	c_{82}	−5
c_{11}	−12313878048	c_{35}	−4173943805202949425295	c_{59}	−3221567119818111932005331	c_{83}	−2
c_{12}	−10202112091	c_{36}	−6680409986407372105125	c_{60}	−3940009595773878157721312	c_{84}	19
c_{13}	293817357505	c_{37}	14450901317815691334394	c_{61}	2517779445482630000908051	c_{85}	2
c_{14}	216724882934	c_{38}	24095889522978307881974	c_{62}	2695738930378788934960920	c_{86}	−
c_{15}	−5682722305632	c_{39}	−43866624565116656397507	c_{63}	−1720776609976061126355295	c_{87}	−
c_{16}	−4173406287546	c_{40}	−75613179506110501552285	c_{64}	−1577767496150339483739065	c_{88}	−
c_{17}	90630479396742	c_{41}	117052843300784351895551	c_{65}	1021699504927039797786209	c_{89}	−
c_{18}	70942431105742	c_{42}	206850256913784340514310	c_{66}	783885981827820754303440	c_{90}	−
c_{19}	−1206873118725580	c_{43}	−275221920209288883000328	c_{67}	−522941711292362568620546	c_{91}	−
c_{20}	−1038627459765548	c_{44}	−494059805036826330310600	c_{68}	−327391707177664571967312	c_{92}	−
c_{21}	13546989348983330	c_{45}	571475852839302185438912	c_{69}	228698352361921804093564	c_{93}	−
c_{22}	12923874392803629	c_{46}	1031331263937081926192731	c_{70}	113498101643535878863400	c_{94}	−
c_{23}	−129141994507456632	c_{47}	−1049963481126561247359686	c_{71}	−84602761720891649834720	c_{95}	−
c_{24}	−136147789009058363	c_{48}	−1882374998260544296663452	c_{72}	−32106025698528772264968	c_{96}	−

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 96, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $96 + 4$ terms removed returns order 96 as well, so $\deg q_{\min} = 96 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $96 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 96 that this sequence satisfies — and that family turns out to have exactly one member.

References

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